

"The Art of Questioning to improve Testing, Agile, and Automating"

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Asking Questions to...

- Build a model: risks, issues, gaps
- What questions expose the risks?
- What questions trigger action?
- What questions foster responsibility?

5 Whys

- Common 'management' questioning
- Toyota
- (Lean) Six Sigma Tool
- Root Cause Analysis

Fritz Perls on Why

“If we spend our time looking for **causes instead of structure** we may as well give up the idea of therapy and join the group of worrying grandmothers who **attack their prey** with such **pointless questions** as “Why did you catch that cold?” “Why have you been so naughty?” ”

Fritz Perls,
The Gestalt Approach

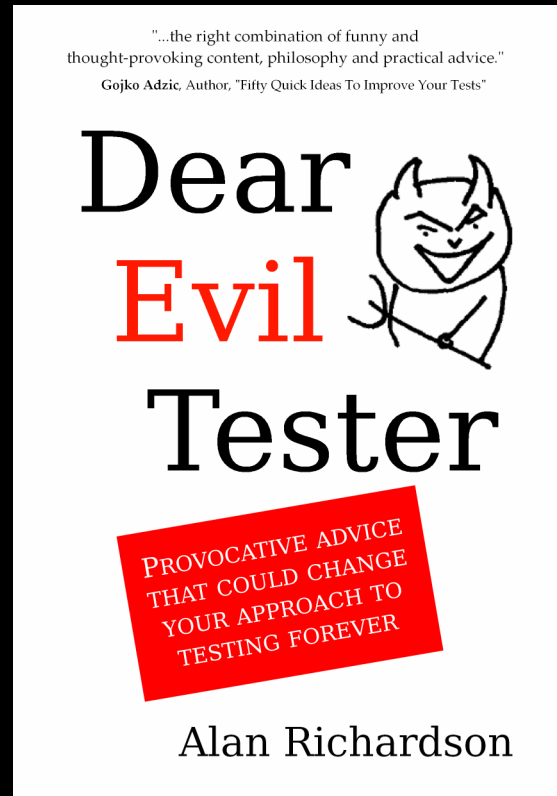
“Why?” is a Belief Question

- Beliefs about causes
- Reinforces a model
 - Justification
- Leads to:
 - Because...

How? What? Where? When? Who?

- Explore a model
- Build a system
- Lead to
 - Experiments
 - Action
 - Responsibility

Dear Evil Tester



Q&A Agony Aunt for Testers

Sometimes we don't know what to
do and just need some advice.

Q: Should I pretend to test?

“In the past I have worked with project managers who have only pretended to manage. Just in case I come across this sort again in the future, I want to know; should I only pretend to test too, as a form of self-preservation?”

Eliza

A: No

Dear Eliza,

...

"NO".

...

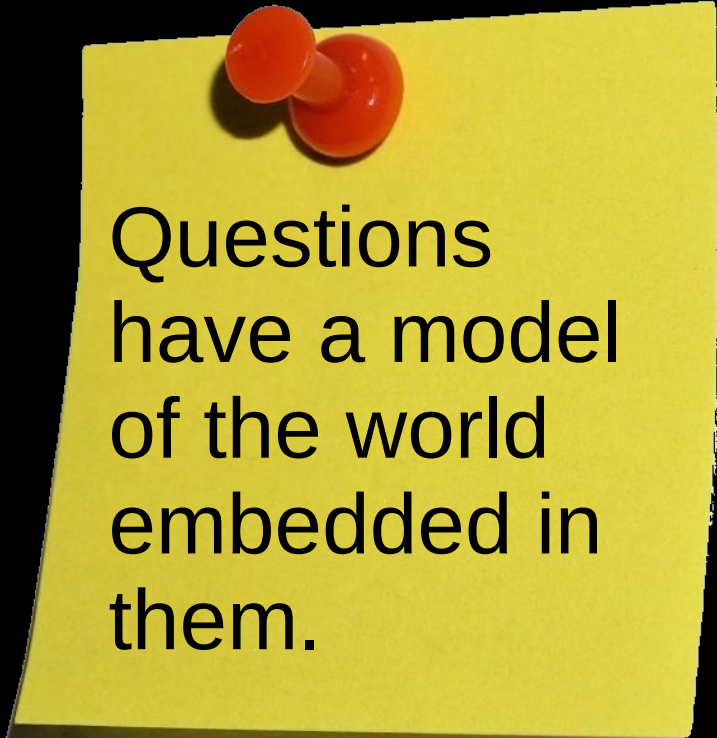
All the best,

Uncle Evil

Q: Should I pretend to test?

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Eliza



Questions
have a model
of the world
embedded in
them.

Q: Tester not pulling their weight?

Dear Evil Tester,

What's the best way to deal
with a fellow tester who is not
pulling his/her weight?

Anon

A: Delegate Upwards

Dear Anon,

...raise your concerns to your manager, after all your **lazy** manager usually has plenty of time on their hands, and it is their responsibility to deal with your light-weight under performing co-workers.

Yours,

Team Spirit Coach Evil

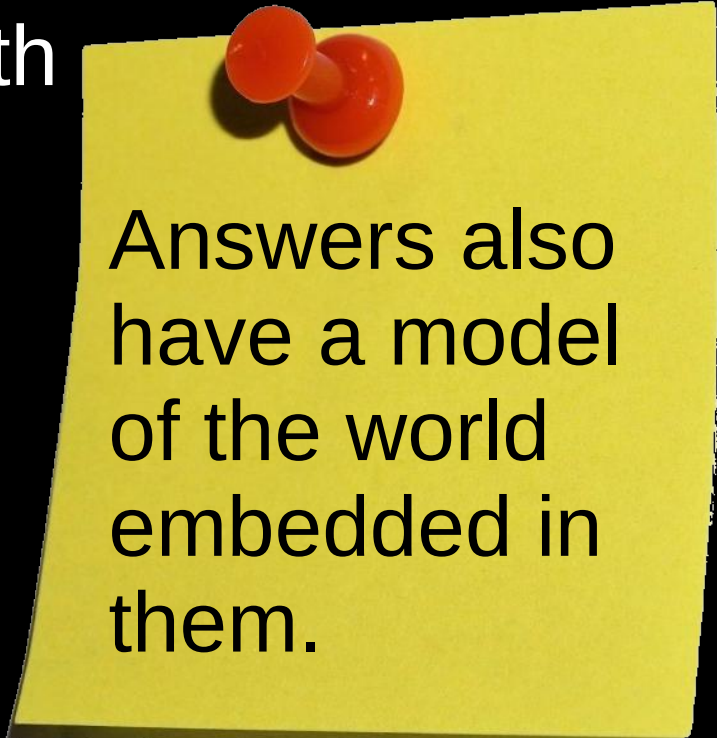
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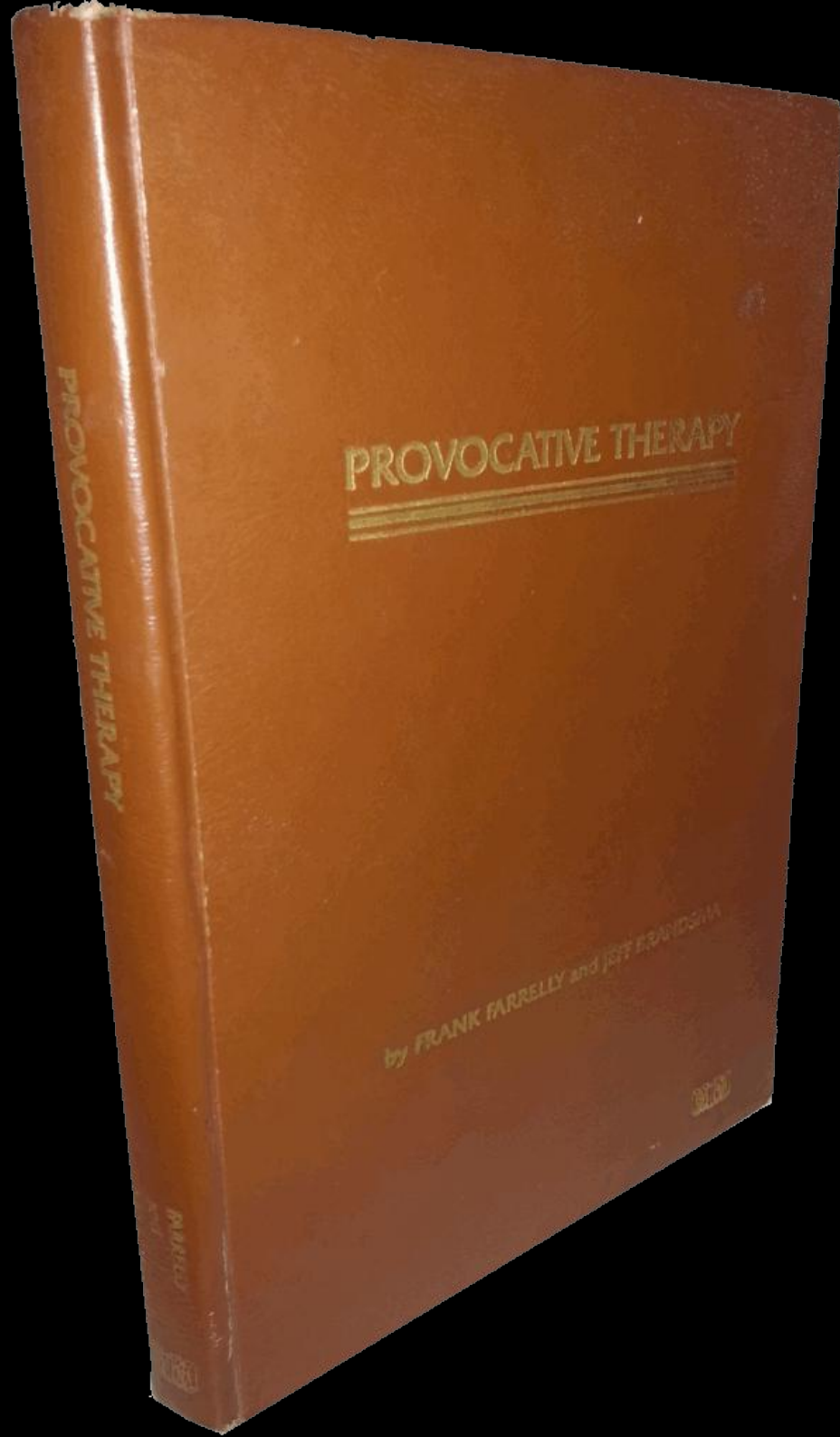
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Team Spirit Coach Evil



Answers also have a model of the world embedded in them.

Q&A Provocation



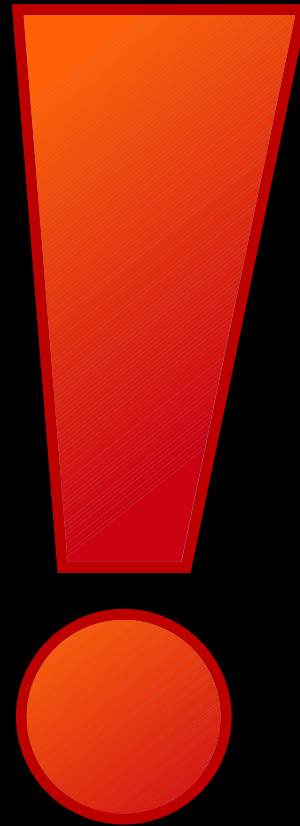
Some Questions are Practical

Q: How to Track Exploratory Testing

Dear Evil Tester,

Do you recommend any tools for note-taking and managing test sessions?

Dependency Danger



What do I do next?

What do I do next again?

Consultancy Job Security

- Job Security – 'the answer person'
- Consultants / Managers / Leads
 - Aim to go 'out of business'
 - build flexibility
 - not dependency

Dependency

- Asking questions for 'answers' rather than to build a model to increase understanding
- A step by step model has no flexibility
- Not learning to ask questions that help when expert is not present

Avoiding Dependency

- Lead by example
 - “asking the right questions”
- Ask questions
 - which expose underlying model
 - to prompt exploration of model
 - Gaps, risks
 - to prompt comparison of model to system
 - Issues, bugs

Modelling Testing as Questioning

- Given a model, does the System match the model?
- Requirements say X, can I do X?
- System 'looks like' I should be able to do Y, can I?

Modelling Testing as Questioning

- Under this model:
 - Test == Ask a Question
 - Answer -> Expand/Confirm Model
 - Reporting == Communicate Model

Modelling Agile as Questioning

- Agile
 - responding to change
 - Change based on learning
 - Learning == changes to models & understanding

Modelling Agile as Questioning

Agile requires asking a lot of questions

- What are we doing?
- Why?
- How Well?
- Could be better?
- How much?
- Minimum acceptable?
- constantly

Automating as Questioning

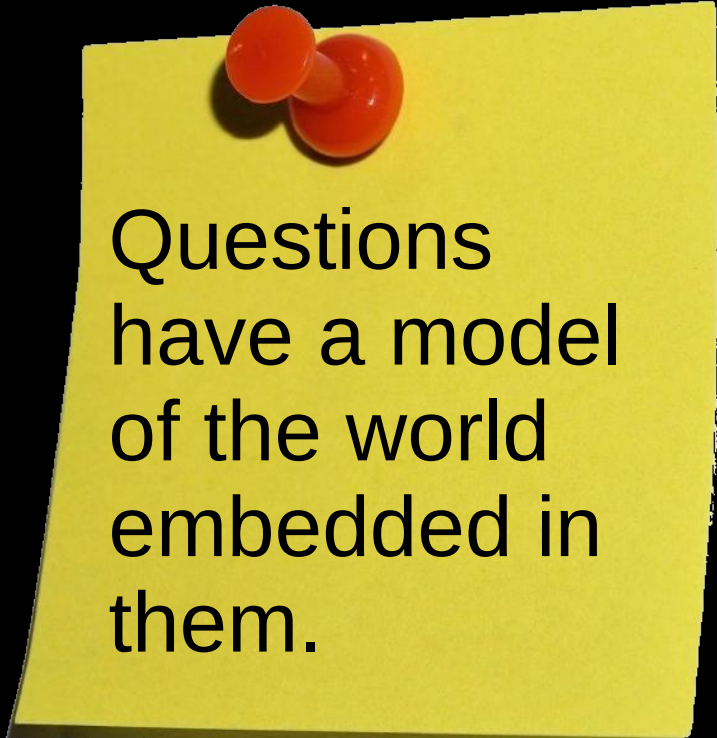
- Automate putting system into a specific state
- Codify specific pre-defined questions
- Assert on the answers

...Repeat

Automating as Questioning

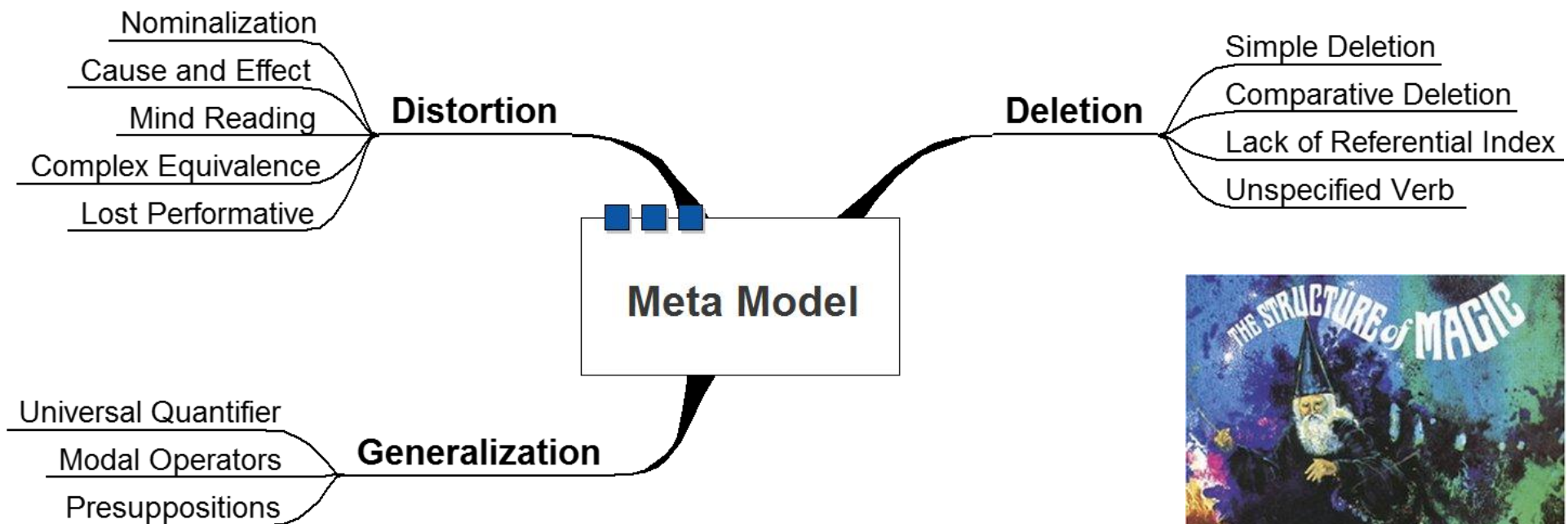
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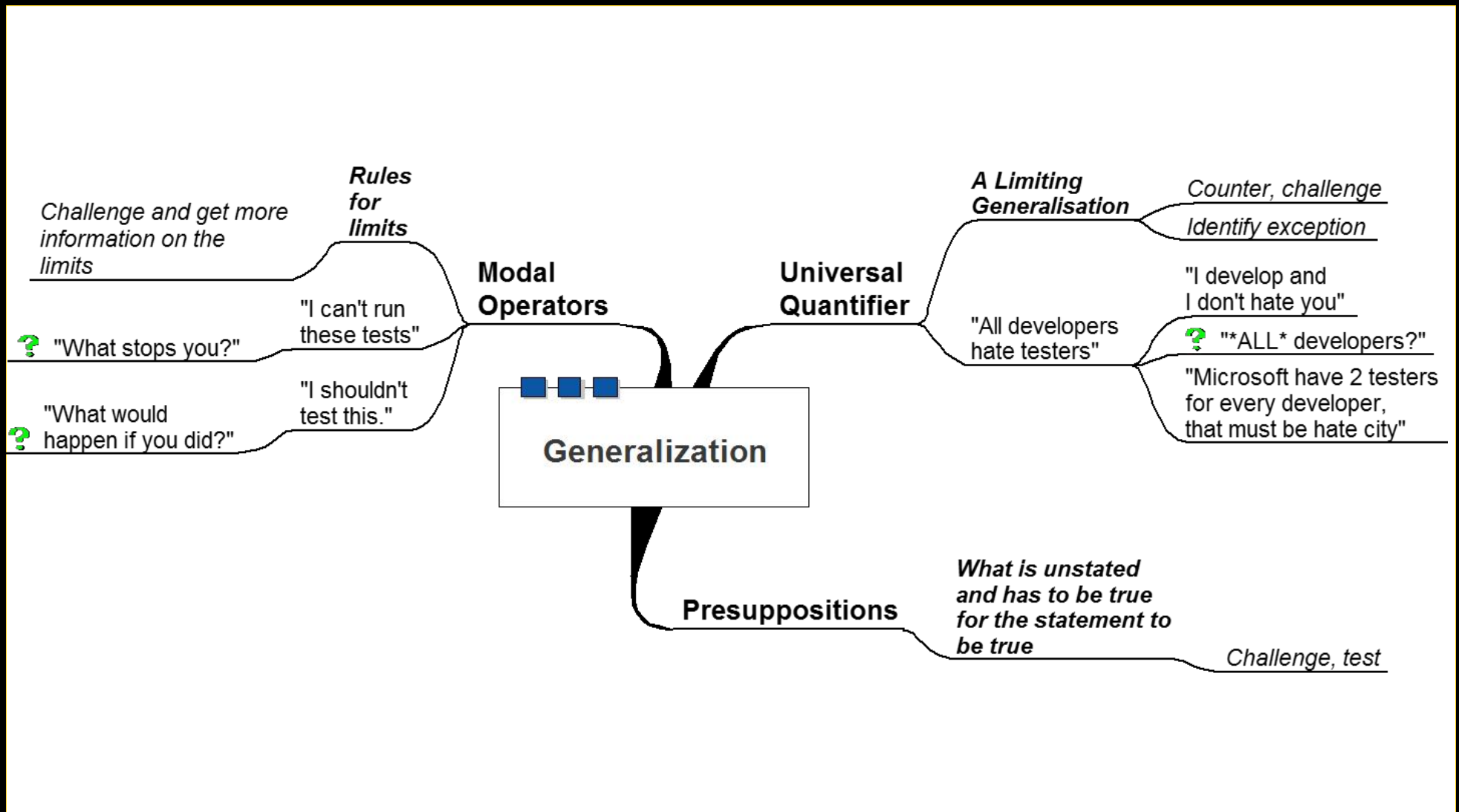


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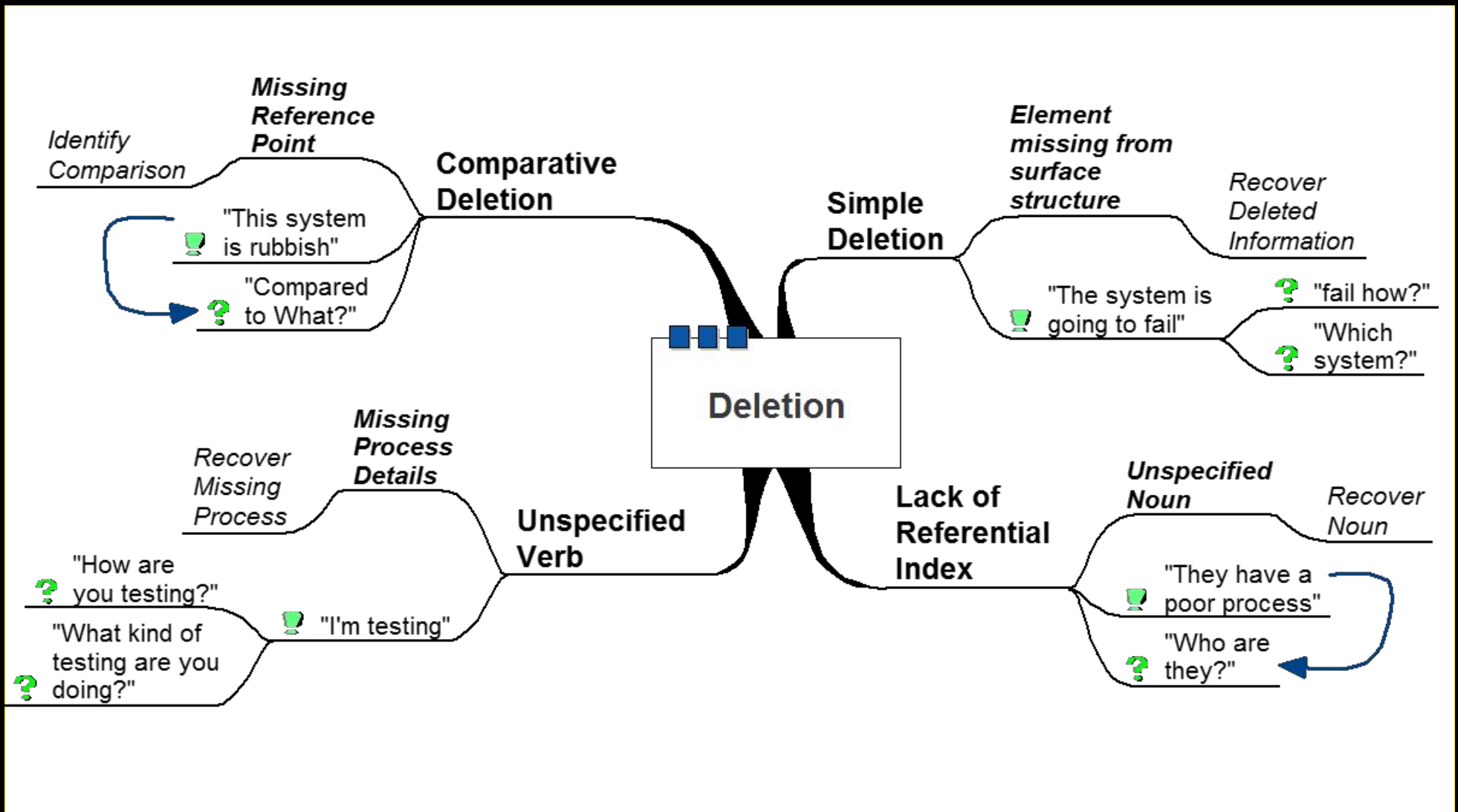
Meta-Model



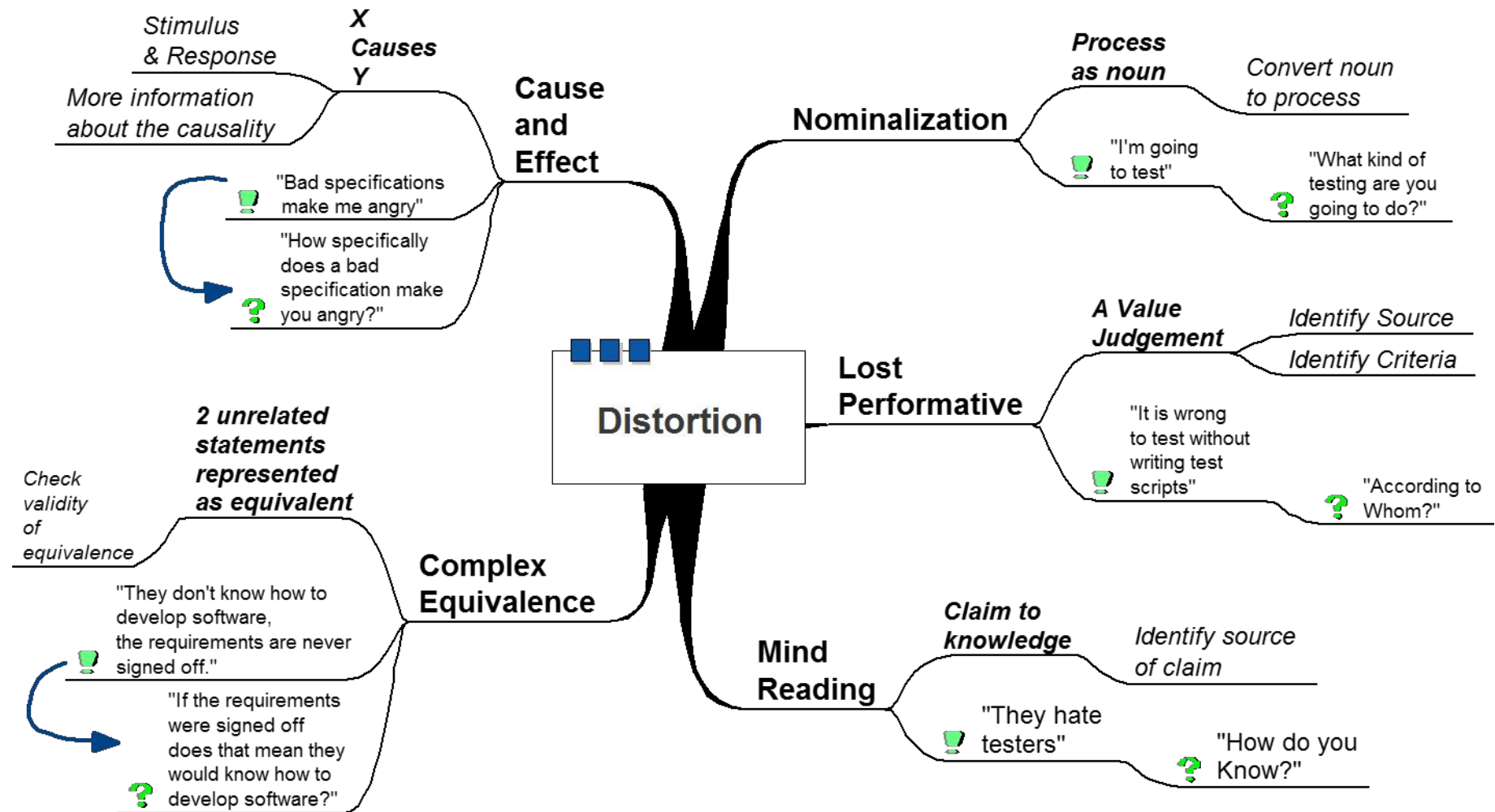
Meta-Model



Meta-Model



Meta-Model



Summary

- The questions we ask reveal our model of the world.
- We can ask questions of 'the world' to explore and expand our model.

Ask The Questions!

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